

Learning Human-Grounded Interaction Strategies for Interactive Agents through Action-Mediated Language Policies

ANONYMOUS AUTHOR(S)

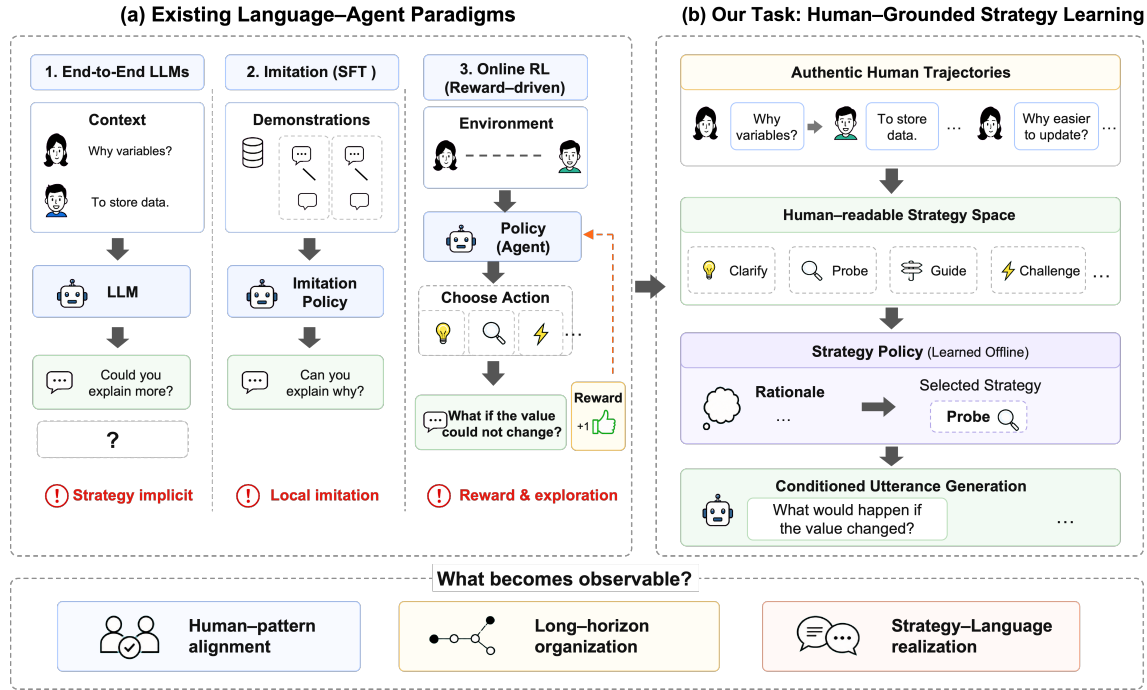


Fig. 1. **Action mediation makes interaction strategy explicit and inspectable.** End-to-end generation leaves strategy implicit, supervised imitation emphasizes local reproduction, and online reinforcement learning depends on reward design and exploration. LA-IQL instead learns offline from authentic human trajectories over a human-readable strategy space, exposes a rationale and selected strategy, and conditions utterance generation on that choice. This explicit boundary makes human-pattern alignment, long-horizon organization, and strategy-language realization separately observable.

Interactive agents are becoming ubiquitous in daily life, from voice assistants and tutoring systems to customer service bots. With this development, dialogue has emerged as the primary channel through which they understand human needs and accomplish tasks. Yet most agents remain narrowly skilled at answering questions while struggling with the richer interactional work that makes human conversation effective: guiding, probing, challenging, building on prior contributions, and shifting initiative across turns. These strategic choices remain implicit in end-to-end language generation, making them difficult to learn from human interaction, inspect, and evaluate. We introduce Language-Action Inverse Q-Learning (LA-IQL), an action-mediated framework that learns offline from human classroom trajectories without online exploration or a hand-designed scalar reward. LA-IQL separates rationale generation, selection over human-readable interaction acts, and language realization, making strategy choice explicitly inspectable. Across long-horizon rollouts on held-out K-12 lessons, the evaluated TD-regularized checkpoint shows substantially less recurrent behavior than its no-TD counterpart while maintaining similar local human-pattern alignment. A metric-matched Markov control further shows that strong local alignment can be achieved without reading the dialogue, while rationale and semantic-intervention analyses indicate that the learned policy captures action-relevant and context-sensitive information. Direct selection more closely matches held-out human action patterns, whereas proposal-constrained re-ranking reduces recurrence, revealing an alignment-repetition trade-off. These

results show that action mediation provides an inspectable interface for separating human-pattern alignment, temporal organization, context sensitivity, and language realization. This can help move agent interaction beyond surface fluency toward more natural, human-grounded interaction.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Reinforcement learning*.

Additional Key Words and Phrases: human-grounded interaction, interactive agents, action-mediated policy, offline imitation learning, dialogue strategy

ACM Reference Format:

Anonymous Author(s). 2027. Learning Human-Grounded Interaction Strategies for Interactive Agents through Action-Mediated Language Policies. In *Proceedings of CHI Conference on Human Factors in Computing Systems (CHI '27)*. ACM, New York, NY, USA, 26 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Introduction

Interactive agents have become woven into the fabric of everyday life. Voice assistants field our questions, tutoring systems guide learners through exercises, customer service bots resolve issues, and companion agents provide social engagement. Across these diverse settings, dialogue serves as the fundamental interface through which agents interpret human intent, negotiate meaning, and accomplish tasks [Park et al. 2023; Xi et al. 2025]. The quality of this dialogic interaction increasingly determines whether these systems feel helpful, natural, and engaging, or mechanical, rigid, and unsatisfying.

Yet a persistent gap separates how current agents converse from how people naturally interact. Most contemporary dialogue agents excel at answering questions and executing direct commands, but struggle with the broader interactional repertoire that characterizes effective human communication. Skilled human interlocutors do far more than respond: they ask clarifying questions, probe incomplete answers, challenge assumptions, offer guidance, build on earlier contributions, and acknowledge and take up what others have said. These interaction-level moves—distinct from the propositional content of any single utterance—shape whether conversations feel dynamic and responsive or flat and repetitive. When agents lack these capacities, interactions become predictable exchanges of information: the agent waits for a question, produces an answer, and waits again. This works for simple queries but fails precisely in the rich, multi-turn, mixed-initiative settings where interactive agents hold the most promise.

The root of this limitation is architectural. Current interactive agents typically generate dialogue end to end, which leaves the interaction strategy behind each response implicit in the generated text [Achiam et al. 2023; Kasneci et al. 2023; Park et al. 2023; Xi et al. 2025]. A locally plausible utterance may clarify, challenge, guide, repeat an earlier pattern, or cede initiative, but these strategic choices are never explicitly represented, evaluated, or controlled. Individually fluent turns can accumulate into repetitive or poorly coordinated interaction [Li et al. 2016a; Zhao et al. 2017]. Behavioral cloning and supervised fine-tuning provide direct ways to imitate observed behavior, but primarily optimize local action or token likelihood without explicitly modeling how interaction choices unfold over time. Online reinforcement learning can optimize sequential behavior, but requires exploration that may be costly or unsafe when interaction involves people or high-stakes environments [Li et al. 2016b; Ouyang et al. 2022; Schulman et al. 2017].

This motivates a central question: *how can interactive agents learn richer, more human-like dialogue strategies from authentic human interaction, while keeping strategy selection separate enough from language realization to be inspected, evaluated, and improved?* We argue that the path forward requires making interaction strategy an explicit, learnable,

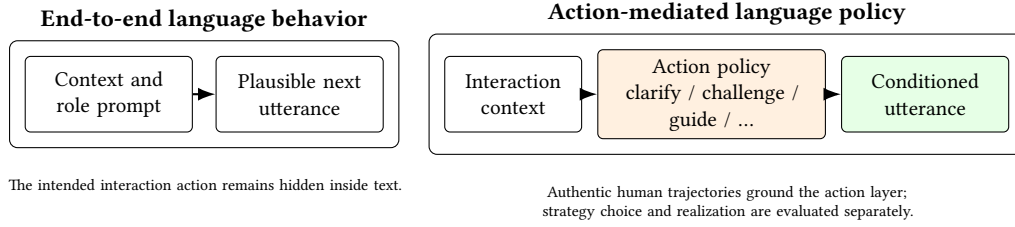


Fig. 2. Human-grounded, action-mediated interaction. LA-IQL learns a human-readable strategy layer from authentic interaction trajectories and separates strategy selection from language realization. This boundary makes behavioral patterns traceable while exposing a realization risk when the final utterance does not express the selected action.

and inspectable object, not a hidden byproduct of text generation. This means separating what move to make—clarify, challenge, guide, or acknowledge—from how to word it, so that each can be learned from human demonstration and evaluated on its own terms.

We study this question through a human-grounded formulation in which the action repertoire, training trajectories, and evaluation references are derived from observed human interaction. Here, *human-grounded* refers specifically to the source of the interaction structure: the action space is defined by an established cognitive-act annotation scheme [Song et al. 2023], and training comes from authentic transcripts. It does not imply that generated dialogue is globally human-like, natural, contextually appropriate, or effective for people.

Our central contribution is an action-mediated paradigm in which a language policy first chooses a discrete, human-readable interaction act and then realizes that choice in language. This separates the source of a strategy—human demonstrations—from the wording of any individual turn. It also creates a traceable boundary at which strategy learning, temporal organization, control allocation, and the eventual language realization can be evaluated as related but distinct objects. Learning from fixed trajectories avoids online exploration and a hand-designed scalar reward [Li et al. 2016b; Ouyang et al. 2022; Schulman et al. 2017].

We investigate this problem through classroom dialogue, a testbed containing frequent, fine-grained interaction moves. Teacher–student exchanges are multi-role and open-ended, combining questioning, uptake, elaboration, challenge, guidance, and synthesis rather than following a fixed task script. Authentic transcripts pair this richness with an established cognitive-act annotation scheme, which provides an observable operationalization of interaction strategies [Song et al. 2023]. Our aim is to study how strategy-rich human interaction can ground an interactive agent’s conversational policy.

We propose **Language-Action Inverse Q-Learning (LA-IQL)**, an offline framework configured by a domain-specific action space and human demonstrations. LA-IQL learns action trajectories with rationale-conditioned offline inverse soft Q-learning over fixed transitions [Garg et al. 2021; Wulfmeier et al. 2024], selects an interaction act from the learned policy, and delegates its language realization to a role model. We evaluate whether generated trajectories reproduce held-out human action patterns, whether that alignment coexists with repetitive behavior, and whether rationales carry action-relevant information. We additionally evaluate how action-control allocation relates to observable knowledge introduction, connection, and development across the resulting long-horizon dialogues. Across 250 evidence-grounded knowledge graphs, Direct selection introduces fewer unique entities than Candidate while retaining nearly identical relational degree and late-stage relation growth. This reveals a more compact rather than uniformly richer knowledge-construction pattern.

Our contributions are threefold:

- An action-mediated formulation for human-grounded interaction-strategy learning that separates explicit strategy selection from language realization, enabling interactive agents to learn not just what to say but how to interact.
- LA-IQL, which combines rationale-conditioned action prediction with offline inverse Q-learning over fixed human interaction trajectories, learning sequential interaction policies without online exploration or hand-crafted rewards.
- A long-horizon evaluation protocol for examining alignment with human action patterns, recurrence, context sensitivity, associations between rationales and actions, language realization, and knowledge structure. This reveals behavioral differences that local similarity or fluent text alone can obscure.

By making interaction strategy explicit and learnable from human demonstration, this work takes a step toward interactive agents that do not merely answer questions, but engage in the rich, adaptive, and genuinely collaborative dialogue that makes human conversation effective.

2 Related Work

2.1 Interaction-Strategy Learning in Pedagogical Dialogue

End-to-end dialogue generation has progressed from sequence-to-sequence and hierarchical models to pretrained Transformers [Serban et al. 2017; Sutskever et al. 2014; Vinyals and Le 2015]. Role prompting and multi-agent simulation now support extended social, task, and classroom interactions [Park et al. 2023; Xi et al. 2025; Zhang et al. 2025], but fluent role realization does not by itself expose or control the pedagogical decision that produced an utterance. Recent educational systems therefore introduce intermediate instructional structure. BIPED derives fine-grained tutor and student dialogue acts from human tutoring data and first predicts a tutor act before generating its response [Kwon et al. 2024]. StratL steers an LLM through a prespecified multi-turn tutoring plan represented as a transition graph [Puech et al. 2025], while ScaffoldLM maintains a stepwise plan, inferred learner state, and assessment-driven memory to select tutoring actions across turns [Li et al. 2026]. Together, these systems show a shift from unconstrained response generation toward explicit pedagogical planning and action selection. LA-IQL shares this separation of decision and realization, but learns a policy over human-annotated cognitive interaction acts from fixed classroom trajectories rather than prescribing a tutoring plan or inferring a task-specific solution path.

This separation also connects educational dialogue to dialogue-policy learning. Task-oriented dialogue is commonly formulated as sequential decision making, with abstract actions selected from dialogue states and policies optimized by reinforcement learning [Li et al. 2016b, 2019]. In educational dialogue, recent online RL aligns tutors through simulated student–tutor interaction and an explicit trade-off between pedagogical support and task success [Dinucu-Jianu et al. 2025]. Such simulation-based optimization offers controllable objectives but depends on the validity of the simulator and reward. Offline alternatives avoid new interaction: CHAI ranks language-model candidates with a learned Q-function [Verma et al. 2022], IQ-Learn recovers a policy from expert demonstrations [Garg et al. 2021], and Wulfmeier et al. [Wulfmeier et al. 2024] cast inverse soft Q-learning for language as MLE with temporal-difference regularization. LA-IQL adapts this last objective from token-level imitation to a 31-class teacher–student action policy (two roles $\times$ 15 cognitive-act symbols + END) grounded in authentic classroom trajectories [Amatari 2015; Song et al. 2023]. Its contribution is not a new IRL objective: it combines rationale-conditioned action selection, action-level TD regularization, and separate language realization without online exploration or a hand-designed scalar reward. This makes pedagogical dialogue an observable testbed for the broader interactive-agent problem of learning how to interact, not only what to say.

2.2 Conversational Agency in Human–AI Interaction

HCI research treats conversational agency as more than response quality: an agent must decide when and how to contribute, how much control to assume, and how its behavior remains legible to people. The Inner Thoughts framework separates covert thought formation from overt participation to support proactive conversation [Liu et al. 2025]. Studies of shared autonomy find non-monotonic trade-offs: delegating communication can aid multitasking while creating discomfort, whereas shared control can impose monitoring costs [Cheng et al. 2025]. In group conversation, users also resist agents that dominate and seek controls over when, what, and where an agent contributes [Houde et al. 2025]. These works frame initiative as an allocation of control rather than a uniformly desirable increase in autonomy. Our allocation boundary is internal to the generated interaction: we compare a learned policy that selects from the full action space with proposal-constrained re-ranking in which the role model first limits the available actions. This does not test user control or establish a preferred autonomy level; it makes the policy–role-model division inspectable at the action layer.

2.3 Evaluating Interactive Behavior in Pedagogical Dialogue

Evaluation must likewise distinguish structural fidelity from pedagogical quality. MRBench provides human annotations over eight learning-science-based dimensions of tutor responses [Maurya et al. 2025]. ReWIRED annotates fine-grained teaching acts in expert explanatory dialogue and shows both that few-shot LLMs struggle to recover these acts and that functional explanation quality requires evaluation beyond static structural metrics [Anagnostopoulou et al. 2025]. EducationQ instead evaluates teaching in dynamic multi-agent scenarios and finds that general model capability does not map monotonically to teaching effectiveness [Shi et al. 2025]. Knowledge-graph representations can provide a semantic layer between dialogue utterances and grounded information items [Schneider et al. 2024], but they do not by themselves establish pedagogical value or learning.

The validity of simulated learners is a separate concern. Teachers report that prompted LLM students can be unnaturally attentive, linguistically complex, and logically inconsistent [Martynova et al. 2025]; a broader 2026 benchmark likewise finds simple prompting inadequate across linguistic, behavioral, and cognitive simulation criteria [Scarlato et al. 2026]. We therefore use classroom simulation as an observable testbed for interactive-agent strategy evaluation, treating action-pattern diversity, expert alignment, rationale–action association, and observable knowledge structures as distinct diagnostics. We do not treat generated classroom trajectories as evidence of authentic learner behavior, instructional appropriateness, or learning outcomes.

3 Method

LA-IQL models an interactive agent as a sequential decision maker: at each step, it observes dialogue history, chooses a domain-defined interaction action, and realizes that action in natural language. Authentic human trajectories ground the action space and supervision, while the explicit action boundary keeps strategy selection inspectable apart from language realization. Following the imitation-learning perspective of offline inverse reinforcement learning, LA-IQL keeps the scalability of supervised fine-tuning while adding a dynamics-aware regularizer over expert trajectories. Our technical contribution is the integration of rationale-conditioned policy states, action-level TD-regularized imitation, and direct policy decoding with separate language realization rather than a new IRL objective. We additionally examine proposal-constrained candidate re-ranking as an inference-time adjustment.

Category	Symbols	Role expansion
Prior-known knowledge	Ipk, Pk	student/teacher $\times$ symbol
Personal information	Ipi, Pi	student/teacher $\times$ symbol
Analysis	Ia, An	student/teacher $\times$ symbol
Coordination	Ic, Co	student/teacher $\times$ symbol
Speculation	Is, Sp	student/teacher $\times$ symbol
Uptake	Iu, Up	student/teacher $\times$ symbol
Agreement, querying, guide	Ag, Qu, Gu	student/teacher $\times$ symbol
END	episode termination	role-independent

Table 1. Compact definition of the 31-action interaction space.

3.1 Problem Formulation and Interaction Action Space

We instantiate the interactive-agent problem as a finite-context Markov decision process (MDP) approximation for offline policy learning from expert classroom demonstrations. Let H_t denote the full interaction history before step t . The policy observation $s_t = f_{\text{short}}(H_t)$ consists of the previous two dialogue turns and two action labels; it is an approximate Markov state that may omit longer-range instructional context. The action a_t is the next interaction act to be generated. It is drawn from a 31-action discrete space: $\{\text{student, teacher}\} \times 15$ cognitive-act symbols (e.g., basic-knowledge questioning/responding, analysis and elaboration, summarization, transfer and innovation, response construction, agreement/challenge/guidance), plus a terminal symbol END. A selected utterance u_t and its action advance the history as $H_{t+1} = H_t \oplus (a_t, u_t)$, after which $s_{t+1} = f_{\text{short}}(H_{t+1})$; END marks episode termination. Thus the deployed transition is jointly induced by the action policy and the language realization model rather than by a_t alone. Given expert demonstrations $\mathcal{D}_E = \{(s_t, a_t, s_{t+1})\}$, we learn an action policy whose generated trajectories are evaluated for diversity and similarity to expert action patterns; language realization is evaluated separately. This action-level formulation is the first LA-IQL design choice: instead of applying inverse RL directly to open-ended tokens, we optimize over human-readable teacher–student cognitive acts and leave surface realization to a separate role model.

We evaluate interaction-pattern diversity (G1), expert-pattern alignment (G2), and rationale–action consistency (G3). G4 targets observable knowledge-construction patterns in generated dialogue through evidence-grounded graph endpoints computed over matched long-horizon trajectories. Together, these goals operationalize the dimensions exposed by the interactive-agent framing: alignment, temporal organization, action-relevant reasoning, language realization, and long-horizon knowledge structure.

3.2 Data Construction and Rationale Annotation

Each expert classroom trajectory is converted into fixed-window transitions. For step t , s_t contains the two most recent utterances and the two most recent action labels, a_t is the annotated next interaction act, and s_{t+1} is obtained after appending the next utterance together with a_t . We use the deepseek-chat API (temperature 0.7) to rewrite rationale targets for 9,168 transitions from the local state and expert next action. The artifact records the model alias and call date but not an immutable provider snapshot. The trained model does not receive the expert action as an inference-time input. It instead learns the factorization

$$p_{\theta}(z_t, a_t | s_t) = p_{\theta}(z_t | s_t) \pi_{\theta}(a_t | s_t, z_t). \quad (1)$$

Rationale targets that copy an action symbol or canonical label are filtered before training to reduce trivial label leakage. Automated generation checks cover length, uniqueness, and retry status; they do not constitute a semantic-quality audit. We use a fixed supervision schema:

<Rationale> explanation text </Rationale> <Action> role-symbol label </Action>.

This construction turns each expert transition into paired supervision for rationale generation and action-level policy learning without a manually specified scalar reward. Because the annotation model sees the expert action whereas the deployed rationale generator does not, training mixes teacher-forced and model-generated rationales to reduce this distribution shift.

3.3 From MLE to Expert Distribution Matching

LA-IQL augments supervised fine-tuning of a rationale-conditioned dual-head model with offline IQLearn. Whereas MLE predicts the observed action locally, the IQLearn term couples the policy to successive state values without online interaction or a learned simulator.

3.4 TD-Regularized Offline IQLearn Objective

We do not introduce a new inverse reinforcement learning objective. Instead, we adopt offline inverse soft Q-learning for language imitation [Wulfmeier et al. 2024]. Grounded in IQ-Learn [Garg et al. 2021], it recasts inverse RL as maximum-likelihood training with a temporal-difference regularizer. Our contribution is its adaptation from token-level generation to a 31-class cognitive-act policy whose Q-values are produced by a rationale-conditioned policy head. We define the augmented policy state as $x_t = (s_t, z_t)$. For teacher-forced examples, $x_t^* = (s_t, z_t^*)$ and $x_{t+1}^* = (s_{t+1}, z_{t+1}^*)$ use the corresponding annotated rationale targets; model-generated-rationale examples replace these targets with samples from $p_\theta(z \mid s)$. At inference, the language head generates z_t from s_t before action scoring. The objective contains four terms. First, $\mathcal{L}_{\text{lm}}$ is the language-modeling loss over the LLM-annotated rationale and action prefix. Second, the policy head is supervised with action-level MLE:

$$\mathcal{L}_{\text{policy}} = -\log \pi_\theta(a_t \mid x_t). \quad (2)$$

Third, the LM head induces a soft action distribution $p_{\text{LM}}(\cdot \mid x_t)$ by scoring the fixed action-label templates after the rationale. We align the policy head with this soft distribution using a KL consistency loss:

$$\mathcal{L}_{\text{consistency}} = D_{\text{KL}}(p_{\text{LM}}(\cdot \mid x_t) \parallel \pi_\theta(\cdot \mid x_t)). \quad (3)$$

Finally, following the formulation of Wulfmeier et al., the policy head produces $Q_\theta(x_t, a)$ for every $a \in \mathcal{A}$. The corresponding soft value and policy are

$$V_\theta(x_t) = \log \sum_{a \in \mathcal{A}} \exp Q_\theta(x_t, a), \quad (4)$$

$$\log \pi_\theta(a \mid x_t) = Q_\theta(x_t, a) - V_\theta(x_t).$$

For an expert transition (x_t, a_t, x_{t+1}) , define the action-level temporal-difference residual:

$$\delta_t := V_\theta(x_t) + \log \pi_\theta(a_t \mid x_t) - \gamma V_\theta(x_{t+1}). \quad (5)$$

We set $V_\theta(x_{t+1}) = 0$ when $a_t = \text{END}$. Since $V_\theta + \log \pi_\theta = Q_\theta$, the corresponding implicit reward is $\delta_t = Q_\theta(x_t, a_t) - \gamma V_\theta(x_{t+1})$. Together, $-\log \pi_\theta(a_t \mid x_t) + \lambda_4 \delta_t^2$ is the action-level counterpart of the MLE-plus-TD objective derived by

Wulfmeier et al. These terms are combined as

$$\mathcal{L} = \mathcal{L}_{\text{lm}} + \lambda_2 \mathcal{L}_{\text{policy}} + \lambda_3 \mathcal{L}_{\text{consistency}} + \lambda_4 \delta_t^2. \quad (6)$$

The coefficient λ_4 is zero in Stages 1–2. In Stage 3 it is set to 1 for Full and remains 0 for no-TD, recovering the non-TD objective when the term is disabled.

Training uses three successive two-epoch LoRA stages. Stage 1 emphasizes structural validity; Stage 2 jointly optimizes language and policy objectives; and Stage 3 continues joint optimization while increasing the proportion of self-generated rationales. The base model is Qwen/Qwen3.5-0.8B, the maximum sequence length is 512, and the physical batch size is 4 with one gradient-accumulation step (effective batch size 4). All three stages use training seed 42 and gradient clipping at 1.0. The evaluated Full and no-TD configurations are matched except for the Stage 3 TD-loss weight, which is 1 for Full and 0 for no-TD. Appendix B records the stage-specific values and the fixed boundary of the archived provenance.

3.5 Rationale-Conditioned Policy Parameterization

The model uses an explicit rationale-conditioned dual-head design, which is the third LA-IQL design choice. The language head (causal LM) generates a `<Rationale>` span before the `<Action>` label; the policy head applies cross-attention pooling over the rationale span and outputs one value per action to parameterize $Q(x, a)$ and $\pi(a \mid x)$. Decisions are conditioned on the generated rationale rather than a single pooled representation.

Concretely, given an input token sequence:

- The LM head produces token-level scores for $\mathcal{L}_{\text{lm}}$ and, by scoring action-prefix templates over the 31 actions, yields a soft LM action distribution.
- The policy head pools the rationale text and outputs policy scores for $\mathcal{L}_{\text{policy}}$.
- A consistency loss $\mathcal{L}_{\text{consistency}}$ aligns the LM action-prefix distribution with the policy logits.

3.6 Direct Policy Decoding and Language Realization

At inference, LA-IQL separates short-context action selection from long-context language realization. Let $h_t = f_{\text{long}}(H_t)$ denote the role model’s longer view of the same history, distinct from the policy observation s_t . The policy model first generates $z_t \sim p_{\theta}(z \mid s_t)$ and directly selects

$$\hat{a}_t = \arg \max_{a \in \mathcal{A}} Q_{\theta}(s_t, z_t, a). \quad (7)$$

The role model then produces one realization $u_t \sim q_{\phi}(u \mid h_t, \hat{a}_t)$. Thus the default LA-IQL inference rule exposes the full action space to the learned policy while using the longer context only to realize its selected action.

The role model is recorded as deepseek-chat for both teacher and student roles. It uses a long window of 30 dialogue turns and five action steps; where sampling applies, it uses a temperature of 0.8 and top- p of 1. The artifact retains the mutable API alias rather than an immutable model snapshot. In the reported evaluation, generation stops at END or the 100-action horizon.

We additionally explored candidate re-ranking as an inference-time adjustment. The role model samples six action-utterance candidates in a fixed schema—three for the teacher and three for the student—and thereby induces $q_{\phi}(\mathcal{C}_t \mid h_t)$

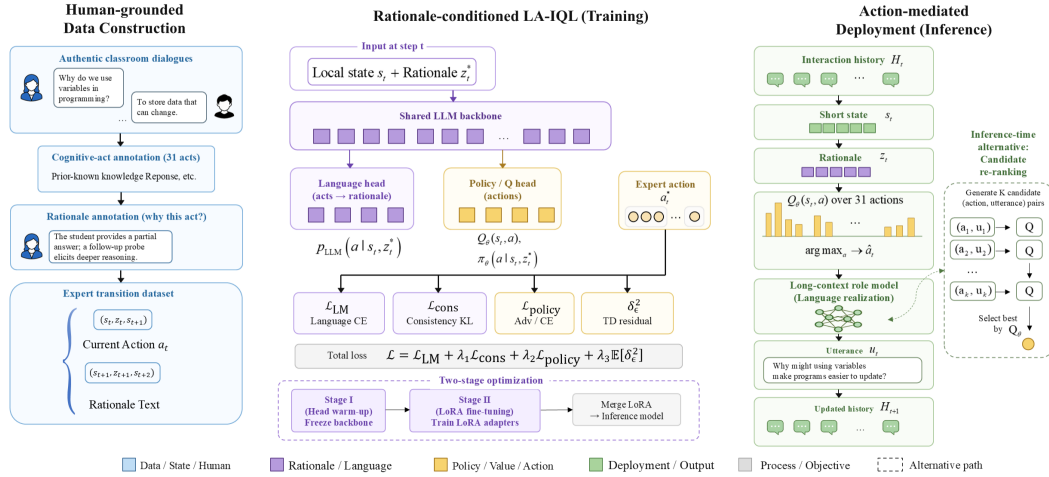


Fig. 3. **LA-IQL framework and technical flow.** Human-grounded classroom dialogue and cognitive-act annotations are converted into rationale-augmented expert transitions (left). Rationale-conditioned LA-IQL jointly trains language and policy/value components with language, consistency, policy, and temporal-difference objectives (center). At inference, the learned policy selects an interaction action and a long-context role model realizes it as an utterance; candidate re-ranking is shown as an inference-time alternative (right).

over $\mathcal{C}_t = \{(a_{t,j}, u_{t,j})\}_{j=1}^6$. The selected index is

$$j^* = \arg \max_{1 \leq j \leq 6} Q_\theta(s_t, z_t, a_{t,j}),$$

$$(a_t, u_t) = (a_{t,j^*}, u_{t,j^*}). \quad (8)$$

This adjustment scores a candidate through its associated action label and does not directly score $u_{t,j}$; candidates sharing an action are therefore indistinguishable to Q_θ , and ties retain proposal order. It is proposal-constrained because an action cannot be selected unless it appears in $\mathcal{C}_t$. Unlike default direct decoding, the role model therefore controls action support while the value model only chooses among proposed actions.

4 Experimental Design

Research questions. We answer four research questions. Together, they test whether an interactive agent’s explicit strategy layer is human-pattern-aligned, temporally organized, context-sensitive, and legibly realized: **RQ1**: How do action-mediated learned policies compare with direct language generation and statistical policies in behavioral diversity, repetition, and alignment with held-out patterns from authentic human interaction? **RQ2**: What long-horizon behavioral differences are observed between the evaluated fixed checkpoints with and without temporal-difference regularization, and between direct full-action-space selection and proposal-constrained candidate re-ranking under the completed configurations? **RQ3**: Do generated rationales contain action-relevant information beyond the dialogue state and explicit label leakage? **RQ4**: What observable patterns of knowledge introduction, connection, and

Component	Recorded configuration
Policy model	Qwen/Qwen3.5-0.8B; 31-action rationale-conditioned dual head
Rationale annotation	deepseek-chat; temperature 0.7; 9,168 rewritten records
Role realization	deepseek-chat; temperature 0.8; top- p 1; 30-turn/5-action context
Policy observation	Two dialogue turns and two action labels; max input length 512; rationale budget 256 tokens
Formal split	35/5/10 lessons; subject-stage stratification; split seed 20260731
Rollout	10 test lessons; horizon 100; END disabled before step 50; margin 3.0
Randomness	One Full checkpoint and one no-TD checkpoint; generation seeds 42, 43, and 44

Table 2. Implementation summary. Values are transcribed from checkpoint, split, annotation, and formal-trajectory artifacts. Historical training fields not retained by those artifacts are documented as unavailable in Appendix B.

development arise under different allocations of action control between the learned policy and the role model across long-horizon simulated classroom dialogues?

4.1 Data and Splits

We use 50 lessons from the K-12 classroom corpus described in Section 3.1. Each dialogue turn is associated with a speaker role and one cognitive-act symbol, from which we construct the 31 interaction acts and the transition tuples used by LA-IQL. We use a frozen, stratified lesson-level split of 35 training, five validation, and 10 test lessons rather than splitting by dialogue turn, so that windows from the same classroom session cannot occur in both partitions. The split is stratified by subject and school stage; lesson identifiers are disjoint across all three partitions, and two lessons used in pilot smoke tests are excluded from the test partition. The split manifest, random seed, data hash, and split-level action frequencies are recorded before formal evaluation. The same test lessons were inspected in an earlier horizon-20 run; we disclose this prior access and do not alter test membership for the horizon-100 baseline evaluation reported below.

4.2 Baselines and Ablations

We compare LA-IQL with five completed controls: direct role-conditioned generation, a constrained teacher prompt, expert-frequency sampling, a first-order Markov policy, and uniform action sampling. The latter three isolate distributional and sequential statistics and are diagnostic controls rather than learned, state-conditioned policies. In particular, the Markov control estimates $p(a_t | a_{t-1})$ from the frozen training split. It is deliberately well matched to bigram- and transition-based G2 measures, but it does not observe dialogue content, the current instructional state, or a rationale, and it does not produce an utterance whose realization can be inspected. Its role is therefore to test how much local human-pattern alignment can be recovered from aggregate action transitions alone. LA-IQL uses the direct decoding rule in Equation 7. We also report *LA-IQL + candidate re-ranking*, an inference-time adjustment that substitutes the proposal-constrained rule in Equation 8 without retraining the policy. This comparison was elevated after inspection of the current held-out results and is therefore descriptive rather than a preregistered model-selection claim. Both configurations use the same checkpoint, lessons, horizon, role-model temperature, and generation seeds. However, the completed Direct trajectories use rationale temperature 0.0 whereas Candidate uses 0.6. We therefore treat them as two system configurations and do not attribute differences solely to decoding mode. To examine the checkpoint-level difference associated with temporal-difference regularization, we add *LA-IQL w/o TD (Direct)*, trained with Stage 3 TD weight 0 rather than 1. It uses Direct decoding, rationale temperature 0.0, the same held-out lessons, role model,

context windows, generation seeds, horizon, and stopping rule as the main LA-IQL system. The recovered training fields otherwise match (Appendix B), but each condition contains one trained checkpoint; we therefore treat this as a fixed-checkpoint contrast rather than a training-seed-general causal estimate.

4.3 Evaluation Metrics

For interaction-pattern diversity (G1), the main analysis reports action Distinct-2, two-step cycle rate, empirical action entropy, and expert-pattern coverage, capturing local variety, repetitive degeneration, distributional diversity, and expert-supported breadth, respectively. For expert-pattern alignment (G2), we use Jensen–Shannon divergence between generated and expert action patterns, transition-matrix distance, and expert support for generated action n -grams. These are behavioral-distribution measures, not tests of whether an action is contextually appropriate for the dialogue state. Their overlap with the Markov control’s inductive bias makes that control a demanding local-pattern reference rather than evidence of state-sensitive interaction competence. For rationale–action consistency (G3), we mask deterministic label strings and train an independent 31-way action predictor on disjoint lessons using the state, the rationale, or both. Our principal automatic measure is incremental simulatability:

$$\Delta_{\text{sim}} = F_1^{\text{macro}}(s_t, z_t \rightarrow a_t) - F_1^{\text{macro}}(s_t \rightarrow a_t), \quad (9)$$

where the target is the evaluated system’s selected action. We also report masked rationale-only Macro-F1, majority, leakage, and shuffled-rationale controls. Predictability alone does not establish causal faithfulness [Hase et al. 2020; Jacovi and Goldberg 2020]; we therefore describe G3 as a simulatability diagnostic rather than an explanation-quality measure. For secondary language-realization diagnostics, we report redundancy and rule-defined severe violations. This deliberately narrow diagnostic set does not measure contextual or classroom appropriateness; open-ended text overlap is secondary because multiple responses can be valid. For G4, the frozen protocol uses system–generation-seed–lesson as the statistical unit. Its primary evidence is an evidence-grounded knowledge-graph extraction followed by deterministic calculation of three prespecified endpoints: unique entities per 100 non-END turns, mean relational degree, and the proportion of final relations first introduced in the latter half of a trajectory. Rule-based diagnostics provide auditable anchors. We report the knowledge-graph endpoints without combining them into a classroom-quality score. Appendix D gives the complete metric and intervention protocol.

4.4 Evaluation Protocol

Action-pattern Jensen–Shannon divergence is the primary G2 automatic endpoint. The reported trajectory results cover generation seeds 42–44 and 10 frozen test lessons per seed at horizon 100. These are three stochastic rollouts from each fixed checkpoint, not independently trained models. We report mean $\pm$ population standard deviation across generation-seed aggregates. For learned-system comparisons, we additionally pair the 30 matching generation-seed–lesson trajectories, cluster resampling by the 10 lessons, run 10,000 bootstrap resamples, and report percentile 95% intervals. The reported TD p -values are unadjusted within an exploratory matched-checkpoint family; the Direct–Candidate table applies Benjamini–Hochberg correction over its 10 retained comparisons. The TD contrast holds Direct decoding and rationale temperature fixed. By contrast, Direct and Candidate differ in rationale temperature and were examined after observing held-out results, so that comparison remains a descriptive system-level contrast rather than a confirmatory causal decoding effect.

Table 3 reports the core G1 metrics.

Method	Dist.-2 $\uparrow$	Cycle-2 $\downarrow$	H_{norm} $\uparrow$	Cov.@20 $\uparrow$
Direct Role LLM	0.4364 $\pm$ 0.0348	0.0286 $\pm$ 0.0088	0.6722 $\pm$ 0.0102	0.5500 $\pm$ 0.0408
Constrained Prompt	0.5279 $\pm$ 0.0117	0.1520 $\pm$ 0.0147	0.6818 $\pm$ 0.0015	0.5333 $\pm$ 0.0236
Uniform Random	0.9593 $\pm$ 0.0048	0.0000 $\pm$ 0.0000	0.9287 $\pm$ 0.0066	0.3500 $\pm$ 0.0000
Expert Frequency	0.6951 $\pm$ 0.0277	0.0067 $\pm$ 0.0033	0.7144 $\pm$ 0.0111	0.9500 $\pm$ 0.0000
First-order Markov	0.5144 $\pm$ 0.0072	0.0819 $\pm$ 0.0140	0.7072 $\pm$ 0.0012	1.0000 $\pm$ 0.0000
LA-IQL w/o TD (Direct)	0.4118 $\pm$ 0.0236	0.2607 $\pm$ 0.0193	0.6412 $\pm$ 0.0120	1.0000 $\pm$ 0.0000
LA-IQL	0.4370 $\pm$ 0.0101	0.1708 $\pm$ 0.0126	0.6670 $\pm$ 0.0084	1.0000 $\pm$ 0.0000
LA-IQL + candidate re-ranking	0.4478 $\pm$ 0.0205	0.0488 $\pm$ 0.0013	0.6664 $\pm$ 0.0131	0.7000 $\pm$ 0.0408

Table 3. Core G1 results (mean $\pm$ population SD over three generation seeds from each fixed checkpoint). Bold values mark the best learned-system result in each column; diagnostic baselines may be diverse without expert alignment.

Method	Paired bigram JS $\downarrow$	Pooled bigram JS $\downarrow$	Transition JS $\downarrow$	Bigram support $\uparrow$	Trigram support $\uparrow$
Direct Role LLM	0.1030 $\pm$ 0.0013	0.3423 $\pm$ 0.0184	0.1708 $\pm$ 0.0049	0.7532 $\pm$ 0.0167	0.4250 $\pm$ 0.0267
Constrained Prompt	0.1096 $\pm$ 0.0016	0.3795 $\pm$ 0.0072	0.1880 $\pm$ 0.0069	0.9218 $\pm$ 0.0092	0.4731 $\pm$ 0.0144
Uniform Random	0.1235 $\pm$ 0.0012	0.4661 $\pm$ 0.0042	0.2293 $\pm$ 0.0028	0.4105 $\pm$ 0.0249	0.0431 $\pm$ 0.0047
Expert Frequency	0.1036 $\pm$ 0.0006	0.2868 $\pm$ 0.0036	0.1995 $\pm$ 0.0036	0.8125 $\pm$ 0.0161	0.3316 $\pm$ 0.0032
First-order Markov	0.0791 $\pm$ 0.0009	0.0982 $\pm$ 0.0038	0.1249 $\pm$ 0.0017	0.9935 $\pm$ 0.0021	0.8557 $\pm$ 0.0169
LA-IQL w/o TD (Direct)	0.0835 $\pm$ 0.0022	0.1530 $\pm$ 0.0177	0.1308 $\pm$ 0.0041	0.9833 $\pm$ 0.0005	0.8773 $\pm$ 0.0125
LA-IQL	0.0827 $\pm$ 0.0031	0.1637 $\pm$ 0.0085	0.1299 $\pm$ 0.0050	0.9906 $\pm$ 0.0024	0.8772 $\pm$ 0.0064
LA-IQL + candidate re-ranking	0.1059 $\pm$ 0.0025	0.3116 $\pm$ 0.0132	0.1648 $\pm$ 0.0035	0.9522 $\pm$ 0.0045	0.4874 $\pm$ 0.0063

Table 4. G2 expert-pattern alignment (mean $\pm$ population SD over three generation seeds from each fixed checkpoint). Bold values mark learned-system results that are numerically best or effectively tied at the shown precision. The first-order Markov control directly models the local transitions measured by several columns but does not condition on dialogue content.

Table 4 reports G2 on the same runs. Paired JS uses the frozen 31^2 bigram vocabulary and Jeffreys smoothing ($\alpha = 0.5$); transition JS weights rows by expert frequency, and support uses training-split n -grams.

4.5 Experimental Results

Data and implementation. The corpus contains 50 lessons, 9,419 turns, 47 topics, and 9,369 transitions. We use the frozen lesson-level split described in Section 4: 35 training, five validation, and 10 test lessons. Each system has three generation runs (seeds 42–44) from the same checkpoint, with up to 100 generated actions for each test lesson. All formal trajectories suppress END before action 50. The frequency and Markov controls use only the frozen training split. Main Direct and candidate re-ranking use the `sft_stage3/final` checkpoint; the w/o-TD fixed-checkpoint contrast uses `sft_stage3_woTD/final`. Direct action selection is the default inference rule, whereas candidate re-ranking is a proposal-constrained inference-time configuration. Main Direct and w/o-TD Direct both use rationale temperature 0.0 and otherwise identical generation settings. Candidate uses temperature 0.6, so its comparison with Direct is not a single-factor decoding ablation. Five of 30 w/o-TD trajectories terminate after 57–95 actions; all 30 main Direct trajectories reach 100 actions. This stopping behavior is an observed policy outcome under the shared rule, not a post-hoc truncation.

Goal	Paired metric (TD – w/o TD)	Difference	95% CI	p
G1	Distinct-2 $\uparrow$	+0.0252	[−0.0282, 0.0688]	0.3186
G1	Cycle-2 $\downarrow$	−0.0900	[−0.1400, −0.0438]	0.0002
G1	Motif rate $\downarrow$	−0.1422	[−0.2013, −0.0889]	0.0002
G1	Normalized entropy $\uparrow$	+0.0258	[−0.0156, 0.0659]	0.2204
G2	Paired bigram JSD $\downarrow$	−0.0007	[−0.0063, 0.0043]	0.8281
G2	Transition JSD $\downarrow$	−0.0009	[−0.0091, 0.0055]	0.8581
G2	Bigram support $\uparrow$	+0.0073	[−0.00001, 0.0140]	0.0510
Secondary	Local trigram redundancy $\downarrow$	−0.00001	[−0.0042, 0.0041]	—

Table 5. Fixed-checkpoint temporal-difference contrast under Direct decoding. Values compare LA-IQL with and without the TD term over the same 30 generation-seed-lesson trajectories, clustered by 10 held-out lessons, with 10,000 bootstrap resamples. Both systems use rationale temperature 0.0 and otherwise identical generation settings. Positive differences mean the TD model is numerically larger; metric arrows indicate the desirable direction.

The evaluated TD checkpoint shows less repetitive long-horizon organization. Table 5 reports the matched Direct-decoding checkpoint contrast. The w/o-TD checkpoint has more two-step cycles (0.2607 vs. 0.1708; TD-w/o-TD difference −0.0900, 95% CI [−0.1400, −0.0438], $p = .0002$) and greater motif recurrence (0.5379 vs. 0.3957; difference −0.1422, 95% CI [−0.2013, −0.0889], $p = .0002$). These are 53% and 36% relative differences, respectively. Distinct-2 and normalized entropy are numerically higher with TD but do not differ reliably. Both systems cover all top-20 expert patterns, indicating that the reduction in recurrence does not coincide with abandoning those common patterns.

Local expert-distribution alignment changes little. Paired bigram JSD is 0.0827 with TD and 0.0835 without TD (difference −0.0007, 95% CI [−0.0063, 0.0043], $p = .8281$); transition JSD is likewise not reliably different. Bigram support shows a borderline descriptive advantage for TD (0.9906 vs. 0.9833, $p = .0510$). For these fixed checkpoints, the observed TD-associated difference is concentrated in recurrent sequence structure rather than local imitation fidelity. Our inference is therefore explicitly bounded to the evaluated checkpoints rather than the training objective across random initializations.

Local transition statistics form a strong but limited reference. Tables 3 and 4 report the formal trajectory measurements. The first-order Markov control obtains the lowest paired bigram JSD (0.0791), pooled bigram JSD (0.0982), and transition-row JSD (0.1249), compared with 0.0827, 0.1637, and 0.1299 for Direct LA-IQL. It also reaches 0.9935 bigram support, while Direct LA-IQL is higher on trigram support (0.8772 vs. 0.8557).

This result follows from a substantive match between model and measurement: the control estimates $p(a_t | a_{t-1})$ from training trajectories, while bigram JSD and transition-row JSD measure the corresponding local action structure. High support and coverage can likewise arise from conservatively resampling training-supported transitions. These are legitimate strengths on the stated G2 endpoints, not an error to discount. However, the control never observes the dialogue text or instructional state and cannot expose whether a locally frequent transition is appropriate in context. We therefore interpret it as evidence that local human-pattern reproduction is achievable with a simple sequential statistic, and not as evidence that the Markov policy is a more capable interactive agent. This distinction motivates reporting G2 alongside state-conditioned decision traces, repetition diagnostics, and action-utterance realization checks.

A frozen intervention provides exploratory evidence of text sensitivity. We evaluated the main Full checkpoint on five pre-existing, evenly spaced states from each of the 10 held-out lessons ($N = 50$). Within each lesson, a cyclic donor replaced utterance semantics while preserving the speaker sequence, number of utterances, and action history.

Goal	Paired metric (Direct – Candidate)	Difference	95% CI	BH q
G1	Distinct-2 $\uparrow$	−0.0108	[−0.0377, 0.0165]	0.5483
G1	Cycle-2 $\downarrow$	+0.1220	[0.0986, 0.1447]	< .0001
G1	Motif rate $\downarrow$	+0.2543	[0.2100, 0.2963]	< .0001
G2	Paired bigram JSD $\downarrow$	−0.0232	[−0.0305, −0.0152]	< .0001
G2	Bigram support $\uparrow$	+0.0384	[0.0283, 0.0492]	< .0001
G2	Trigram support $\uparrow$	+0.3898	[0.3551, 0.4238]	< .0001

Table 6. Paired Direct–Candidate comparison over 30 matched generation–seed–lesson units (10,000 bootstrap resamples; percentile intervals; BH correction over 10 retained comparisons). Positive values mean Direct is numerically larger; metric arrows indicate the desirable direction. The configurations also differ in rationale temperature, so this is a system-level comparison rather than a single-factor decoding ablation.

Relative to the original states, the probability assigned to the expert action decreased by 0.0767 (lesson-cluster bootstrap 95% CI [0.0153, 0.1288]), 60% of argmax actions flipped (CI [0.48, 0.72]), and mean policy JSD was 0.2957 (CI [0.2476, 0.3431]). The 31-class Macro-F1 decrease was 0.0201, with an interval crossing zero ([−0.0185, 0.0649]). This intervention indicates that the fixed policy distribution responds to dialogue text beyond the preserved action history. Its small sample, deterministic donor construction, and lack of human judgments mean it does not establish that the response is contextually appropriate. Appendix C records the protocol.

Within the learned architecture, alignment comes with repetitive local structure. Direct LA-IQL is closer to the held-out expert distribution than Candidate: paired bigram JSD is 0.0827 versus 0.1059, and training-expert bigram support is 0.9906 versus 0.9522. Across 30 matched generation–seed–lesson units, the Direct–Candidate difference in bigram JSD is −0.0232, 95% CI [−0.0305, −0.0152], BH-adjusted $q < .0001$.

This alignment does not imply uniformly better interaction dynamics. Direct has substantially more two-step cycles (0.1708 vs. 0.0488; paired difference +0.1220, 95% CI [0.0986, 0.1447], $q < .0001$) and repeated motifs (0.3957 vs. 0.1413; difference +0.2543, 95% CI [0.2100, 0.2963], $q < .0001$). Distinct-2 has no reliable paired difference (−0.0108, 95% CI [−0.0377, 0.0165], $q = .5483$). The configurations therefore differ more clearly in action-sequence organization than in local combinatorial diversity.

A rule-selected trace makes the recurrence difference inspectable. Before reading trajectory text, we selected the matched lesson–seed unit with the largest Direct–Candidate Cycle-2 difference, breaking any tie by motif-rate difference and then lexical order. This rule selects the Grade-9 chemistry lesson on atomic structure at generation seed 43 (Cycle-2: 0.309 vs. 0.052; motif rate: 0.55 vs. 0.17). Table 7 shows the earliest four-turn Direct cycle and the same turn positions in the matched Candidate trajectory. Rationales and utterances are abridged translations from Chinese; action symbols follow Table 1.

The action trace exposes recurring decision types that would be difficult to audit from surface text alone. At the same time, the Direct utterances advance from a claim about nuclear force to the countervailing role of electrostatic repulsion. The Candidate trace also exposes a local rationale–realization shift at t15: the recorded rationale focuses on refining placement and mass, while the utterance asks about electrostatic stability. The example therefore illustrates, but does not by itself validate, the aggregate recurrence contrast or realization quality: a repeated action motif is not equivalent to verbatim repetition or instructional failure. We consequently use the case for structural interpretation rather than a human-preference ranking.

Configuration and local context	Four-turn decision trace: action; rationale → utterance
Direct	
Student had just restated the teacher's explanation of nuclear force.	t13 Teacher-Ag: confirm understanding and extend the issue → "Exactly. Does having more protons mean stronger nuclear force and a more stable nucleus?" t14 Student-Pk: answer the hypothesis from prior knowledge → "I think so; more protons should mean more nuclear force." t15 Teacher-Ag: affirm the analogy, then introduce a competing force → "But electrostatic repulsion also increases; which force wins?" t16 Student-Pk: respond to that comparison → "Repulsion also grows, so the nucleus may not necessarily become more stable."
Candidate	
Student had just estimated electron mass and located most atomic mass in the nucleus.	t13 Teacher-Gu: confirm the estimate and consolidate mass-distribution understanding → "Draw an atomic model and mark each part's position and approximate mass share." t14 Student-An: construct and explain the requested model → "A small central nucleus contains protons and neutrons; light electrons move outside it." t15 Teacher-Iu: refine the diagram's component placement and mass analogy → "What force keeps the electrons and nucleus in a stable relation?" t16 Student-Pk: supply the requested relation → "The attraction between negative electrons and the positive nucleus."

Table 7. Rule-selected inspectable case from formal trajectories (lesson 4873, generation seed 43, turns 13–16). The Direct window instantiates an Ag-Pk-Ag-Pk two-step cycle, whereas Candidate traverses Gu-An-Iu-Pk. The entries preserve the recorded action, rationale intent, and utterance content while abbreviating the original text for space.

Rationales carry action-predictive information. Under the same G3-v2 dual TF-IDF evaluator, masked rationale-only Macro-F1 is 0.3986 for Direct and 0.1013 for Candidate, compared with state-only scores of 0.0159 and 0.0123. Generation-seed-level incremental simulatability Δ_{sim} is 0.3885 for Direct and 0.0863 for Candidate. The w/o-TD Direct checkpoint obtains rationale-only Macro-F1 0.3682, combined Macro-F1 0.3629, $\Delta_{\text{sim}} = 0.3372$, and accuracy 0.6867, descriptively below the corresponding TD Direct values of 0.3986, 0.4044, 0.3885, and 0.7097. For the original Direct and Candidate configurations, the observed generation-seed-level delta exceeds all 20 within-lesson/role shuffle controls, and all 30 generation-seed-lesson deltas are positive. These results show that masked rationales contain information about the selected action. They do not show that rationales causally determine the policy decision or constitute faithful explanations; we also do not attach an inferential TD-effect claim to the descriptive G3 difference.

Rule-based diagnostics delimit language-realization risk. The final language-realization evidence consists of conservative, reproducible rule-based checks. They show no exact repetitions or rule-defined severe violations for Direct, Candidate, or w/o-TD Direct, while local trigram redundancy does not differ reliably between Direct and Candidate or between the two Direct checkpoints. These narrow checks do not measure realization quality or classroom appropriateness.

RQ4 reveals compact but similarly connected knowledge construction. All 250 extracted graphs passed the machine-checkable schema and utterance-ID evidence constraints. Table 8 reports the three prespecified endpoints after first averaging generation seeds within each system-lesson unit.

Relative to Candidate, Direct introduces 7.19 fewer unique entities per 100 turns (95% CI $[-10.43, -4.25]$, Holm-adjusted $p = .0059$), while mean relational degree (-0.003 , CI $[-0.098, 0.093]$) and late-stage relation growth (-0.001 , CI $[-0.065, 0.052]$) are nearly unchanged. Relative to the matched w/o-TD Direct checkpoint, Direct is lower in UER₁₀₀ by 4.00 (CI $[-8.69, -0.05]$, adjusted $p = .2109$), higher in mean relational degree by 0.149 (CI $[0.059, 0.240]$, adjusted $p = .0527$), and indistinguishable in late-stage growth (-0.001 , CI $[-0.048, 0.050]$). Thus, the clearest RQ4 pattern is not uniformly greater knowledge volume: Direct forms a more compact graph than Candidate while preserving

System	UER ₁₀₀	Mean relational degree	Late-stage relation growth
Expert transcript	14.17	1.906	0.456
LA-IQL Direct	23.95	2.112	0.379
LA-IQL Candidate	31.14	2.116	0.379
LA-IQL w/o TD (Direct)	27.96	1.963	0.380
Direct-role LLM	38.65	2.159	0.334
Constrained teacher prompt	39.15	2.164	0.360
Frozen BC	30.36	2.012	0.301
LoRA policy	32.22	2.064	0.325
Likelihood rerank	28.32	2.202	0.335

Table 8. Knowledge-graph endpoints for RQ4. UER₁₀₀ is the number of evidence-grounded unique entities per 100 non-END turns. Generated-system values first average the three generation seeds within each of the 10 held-out lessons; the expert row contains one transcript per lesson. These endpoints describe generated knowledge structure, not factuality or instructional quality.

comparable connectivity and the timing of relation development. UER₁₀₀ has no intrinsic “higher is better” direction, so this result should not be interpreted as an educational-quality ranking.

Appendix A gives the full system-level comparison, G3-v2 table, and judgment status.

5 Discussion

5.1 Human Grounding as an Inspectable Design Resource

Our first contribution is methodological: action mediation renders human grounding an operational and inspectable design resource for building more natural interactive agents. The action vocabulary, training demonstrations, and held-out comparison patterns are all drawn from authentic classroom interaction. It links policy learning, generation, and evaluation through a shared human-readable strategy layer. This goes beyond structured prompting for next-utterance generation: it establishes a stable interface at which designers can inspect what strategy was selected, how strategies unfold over time, and how each choice is realized in language. For HCI practitioners seeking to move beyond agents that merely answer questions toward agents that guide, probe, and collaborate, such traceability is essential. This is because it distinguishes between hoping a model has acquired sound interactional habits and verifying what it has actually learned.

The metric-matched Markov control clarifies what human-pattern alignment can and cannot establish. Its strong local alignment indicates that transition similarity can be reproduced without reading the dialogue; the semantic intervention, by contrast, shows that the learned policy distribution responds to dialogue content beyond the preserved action history. Human-pattern similarity is therefore informative only when situated within a broader, inspectable evaluation rather than treated as a sufficient measure of context-sensitive interaction. This finding carries a cautionary implication for the broader enterprise of learning from human demonstrations: matching surface statistics of human behavior does not guarantee that the agent has internalized the underlying interactional competence. Human grounding still does not ensure globally human-like or contextually appropriate behavior, but the action boundary makes the relevant successes and failures empirically separable, and therefore addressable.

5.2 TD-Associated Differences Reveal Longer-Horizon Organization

The matched Direct-decoding checkpoint contrast isolates the clearest observed TD-associated difference. The w/o-TD checkpoint shows no reliably detected difference in local expert-pattern alignment yet exhibits 53% more two-step cycles and 36% more repeated motifs. This pattern is consistent with the TD term affecting recurrent sequence organization rather than merely expanding the action vocabulary or improving one-step imitation, though the single checkpoint per condition precludes a training-objective-level causal conclusion. The w/o-TD policy also terminates five of 30 trajectories before the 100-action horizon, whereas the TD policy terminates none, providing further descriptive evidence of divergent rollout behavior. Because each condition contains one trained checkpoint, the bootstrap captures held-out lesson and generation variation but not training-seed uncertainty.

This result demonstrates the analytical leverage of the action-mediated formulation for designing less repetitive and more engaging interactive agents. It localizes the observed TD-associated difference in temporal organization (i.e., 35% fewer two-step cycles and 26% fewer repeated motifs for the evaluated TD checkpoint) rather than in local expert-pattern alignment or action-vocabulary expansion. This distinction bears directly on user experience: an agent that cycles through a small repertoire of interaction moves, however locally appropriate, will feel monotonous and predictable in sustained use. End-to-end text evaluation would obscure this distinction, since surface-level lexical variation can mask underlying interactional rigidity. We do not extend this structural result to participants' perceived coherence or usefulness, which lies outside the empirical scope of this paper and constitutes important future work for understanding how these structural properties translate into user experience.

5.3 Action Mediation Makes Control Allocation Observable

Separating strategy selection from language realization turns control allocation into an observable design variable. The completed Direct configuration grants the learned policy access to the full action space and yields stronger alignment with held-out human action patterns, but also produces more short cycles and repeated motifs. The Candidate configuration restricts action support to role-model proposals and exhibits fewer repetitions, yet its trajectories diverge further from the held-out action distribution. Because the configurations also differ in rationale temperature, this constitutes a descriptive system-level contrast. Nevertheless, the completed configurations illustrate that greater policy control over action support need not coincide with lower recurrence: strategy support, temporal organization, and wording must be considered jointly.

This finding has practical implications for how designers configure interactive agents. The trade-off between alignment and repetition—between faithfully reproducing human action distributions and avoiding conversational ruts—cannot be resolved by simply granting the policy more or less control. Rather, it requires careful consideration of where in the generation pipeline different constraints should operate. The explicit action boundary provides a concrete substrate for such structural traceability. It enables system designers to compare the strategy selected for a state with the utterance ultimately produced, inspect recurring control patterns, and localize whether a divergence originates in policy choice or realization. We establish this as a method-level traceability capability. Nevertheless, for HCI researchers and practitioners building interactive systems, this form of inspectability is a prerequisite for responsible iteration and improvement.

5.4 Implications for Evaluating Interactive Dialogue Agents

Three implications follow from the present testbed for evaluating interactive agents that aim to engage in richer, more human-like dialogue. First, action-pattern metrics should be paired with controls whose inductive biases match those metrics; otherwise, local statistical reproduction may be mistaken for context-sensitive competence. An agent that reproduces the local distribution of human interaction moves may still fail to respond meaningfully to what has actually been said, a failure that becomes visible only when one intervenes on the dialogue content itself. Second, evaluation should treat strategy choice and language realization as separate but coupled objects: an appropriate action label is of little use when the utterance fails to express it. Whether the agent intends to guide, challenge, or acknowledge and whether it actually does so through language is an empirical question, not a given. Third, long-horizon repetition must be reported alongside diversity and alignment, because a policy can cover human-supported patterns while repeatedly cycling among a small subset. For interactive agents deployed in sustained use, this is precisely the kind of failure that erodes user trust and engagement even when individual responses appear appropriate.

Taken together, these findings suggest an evaluation framework for strategy-rich interactive agents: expose a readable strategy layer, assess whether the policy uses interaction context, and evaluate human-pattern alignment, temporal organization, and language realization as complementary rather than interchangeable properties. This framework speaks directly to the HCI community's interest in designing systems that are not merely capable but genuinely usable and engaging in real human interaction. The action-mediated approach provides the transparency necessary to shift evaluation from the question of whether an agent produces fluent responses toward whether it interacts effectively and whether the basis for that interaction is observable.

6 Conclusion

Interactive agents are increasingly embedded in daily life, and dialogue remains the primary channel through which they serve human needs. Yet the gap between answering questions well and engaging in genuinely collaborative, adaptive interaction remains wide. This paper introduced an action-mediated formulation for learning human-grounded interaction strategies and instantiated it in LA-IQL. From fixed human interaction trajectories, LA-IQL learns an explicit policy over readable interaction acts through rationale-conditioned action prediction and offline inverse Q-learning, while keeping strategy selection separate from language realization. This formulation makes interaction decisions, temporal organization, and surface realization independently inspectable. This is a necessary condition for building agents that move beyond mechanical responsiveness toward natural, context-sensitive engagement.

Across held-out long-horizon classroom rollouts, the evaluated TD-regularized checkpoint produced less recurrent behavior while preserving similar local human-pattern alignment. Varying how control is allocated between the policy and the role model further exposed a clear trade-off: direct selection better preserved held-out human patterns, whereas proposal-constrained re-ranking reduced recurrence. Together, these findings demonstrate that action mediation reveals consequential differences in interaction organization that fluent text or local similarity alone can obscure.

The broader significance of this work lies in its treatment of human interaction as a learnable and inspectable resource. By making strategy explicit, we enable designers and researchers to pose sharper questions about what interactive agents have actually learned from human demonstration, how they organize behavior over time, and where their limitations reside. The evidence supports an inspectable modeling and evaluation interface for strategy-rich interactive agents. It is bounded to the evaluated checkpoints and structural behavior. These boundaries define the trajectory of future work: extending action-mediated learning to diverse interactional domains, validating structural properties

against user experience, and ultimately building interactive agents that do not merely respond to people, but engage with them.

Limitations

LA-IQL relies on a fixed interaction-act taxonomy derived from one classroom annotation scheme. The resulting 31-action space may omit pedagogically meaningful behaviors and may not transfer unchanged across subjects, grade levels, languages, cultures, or classroom formats. Although the framework is designed around domain-specific state and action spaces and is motivated by a broader interaction-learning problem, this paper evaluates only its classroom-dialogue instantiation and makes no empirical transfer claim to other interactive-agent domains, including conversational assistants, multi-agent systems, or embodied agents. Because the method learns from offline expert demonstrations, it can reproduce annotation errors and systematic biases in the source corpus and cannot identify effective strategies that are absent from that corpus. The finite-context policy observation may miss long-range instructional goals even though the role model receives a longer dialogue history.

The completed Direct and Candidate configurations differ in rationale sampling temperature in addition to their action-selection mechanism. Their comparison therefore identifies a reproducible system-level trade-off but does not isolate the causal effect of decoding mode; decoding-specific causal attribution is outside the claims of this paper.

Generated rationales should not be interpreted as faithful causal explanations of the model’s internal computation. They are supervised textual intermediates and can be plausible while remaining incomplete or inconsistent with the selected action. The explored proposal-constrained adjustment is limited by candidate coverage: a strong Q-function cannot select an action absent from the proposal set, and it does not distinguish utterances sharing the same action label. Direct decoding avoids this support restriction but delegates realization of its selected action to a separate role model, which may fail to express that action faithfully. Finally, automatic action-distribution metrics do not establish educational effectiveness, and this paper makes no claims about learning outcomes or teacher-training utility. The G4 knowledge-graph endpoints likewise measure structures expressed in generated text, not changes in student cognition, factual correctness, instructional quality, or learning outcomes. Machine-checkable schema and utterance references do not by themselves establish that extracted entities and relations are semantically correct, and exact edge identities may vary across repeated automatic extraction. Moreover, the current G2 endpoints emphasize local action n -grams and transition distributions. A policy such as the first-order Markov control can score highly by conservatively reusing training-supported transitions without responding to the semantic content of the current classroom state. Our 50-state intervention provides evidence that the Full policy distribution changes with utterance semantics, but its deterministic donor and small sample bound the conclusion to distributional sensitivity rather than contextual decision quality.

Ethical Considerations

Classroom dialogue can contain personal information and involve minors, making privacy, consent, and data governance central to this work. Any use of classroom transcripts should follow the source dataset’s consent conditions, license, and access restrictions, and should remove identifying information before model training or release. Generated dialogue must not be presented as a substitute for qualified teachers or used for high-stakes assessment without human oversight.

An interactive agent built with this framework may reproduce cultural, linguistic, or pedagogical biases present in the demonstrations and in the underlying language models. In particular, a policy trained on a narrow classroom population may incorrectly treat one discourse style as universally expert. Deployment should therefore include subgroup

and cross-context evaluation, educator review, clear disclosure that outputs are machine-generated, and mechanisms for users to contest or override generated recommendations. LLM-generated rationale annotations also require quality auditing because fluent explanations can conceal labeling errors or unsupported assumptions.

References

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).
- Veronica Odiri Amatori. 2015. The Instructional Process: A Review of Flanders’ Interaction Analysis in a Classroom Setting. *International Journal of Secondary Education* 3, 5 (2015), 43–49. [arXiv:https://article.sciencepublishinggroup.com/pdf/10.11648/j.ijsedu.20150305.11](https://article.sciencepublishinggroup.com/pdf/10.11648/j.ijsedu.20150305.11) doi:10.11648/j.ijsedu.20150305.11
- Aliki Anagnostopoulou, Nils Feldhus, Yi-Sheng Hsu, Milad Alshomary, Henning Wachsmuth, and Daniel Sonntag. 2025. Human and LLM-based Assessment of Teaching Acts in Expert-led Explanatory Dialogues. In *Proceedings of the 6th Workshop on Computational Approaches to Discourse, Context and Document-Level Inferences (CODI 2025)*. Association for Computational Linguistics, Suzhou, China, 166–181. doi:10.18653/v1/2025.codi-1.15
- Yi Fei Cheng, Hirokazu Shirado, and Shunichi Kasahara. 2025. Conversational Agents on Your Behalf: Opportunities and Challenges of Shared Autonomy in Voice Communication for Multitasking. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA. doi:10.1145/3706598.3714017
- Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C. Wallace. 2020. ERASER: A Benchmark to Evaluate Rationalized NLP Models. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 4443–4458. doi:10.18653/v1/2020.acl-main.408
- David Dinucu-Jianu, Jakub Macina, Nico Daheim, Ido Hakimi, Iryna Gurevych, and Mrinmaya Sachan. 2025. From Problem-Solving to Teaching Problem-Solving: Aligning LLMs with Pedagogy Using Reinforcement Learning. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Suzhou, China, 272–292. doi:10.18653/v1/2025.emnlp-main.15
- Divyansh Garg, Shuvam Chakraborty, Chris Cundy, Jiaming Song, and Stefano Ermon. 2021. IQ-Learn: Inverse soft-Q Learning for Imitation. In *Advances in Neural Information Processing Systems*.
- Peter Hase, Shiyue Zhang, Harry Xie, and Mohit Bansal. 2020. Leakage-Adjusted Simulatability: Can Models Generate Non-Trivial Explanations of Their Behavior in Natural Language?. In *Findings of the Association for Computational Linguistics: EMNLP 2020*. Association for Computational Linguistics, 4351–4367. doi:10.18653/v1/2020.findings-emnlp.390
- Stephanie Houde, Kristina Brimjoin, Michael J. Muller, Steven I. Ross, Dario Andrés Silva Moran, Gabriel Enrique Gonzalez, Siya Kunde, Morgan A. Foreman, and Justin D. Weisz. 2025. Controlling AI Agent Participation in Group Conversations: A Human-Centered Approach. In *Proceedings of the 30th International Conference on Intelligent User Interfaces*. Association for Computing Machinery, Cagliari, Italy, 390–408. doi:10.1145/3708359.3712089
- Alon Jacovi and Yoav Goldberg. 2020. Towards Faithfully Interpretable NLP Systems: How Should We Define and Evaluate Faithfulness?. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 4198–4205. doi:10.18653/v1/2020.acl-main.386
- Enkelejda Kasneci, Kathrin Seßler, Stefan Küchemann, Maria Bannert, Daryna Dementieva, Frank Fischer, Urs Gasser, Georg Groh, Stephan Günemann, Eyke Hüllermeier, et al. 2023. ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and individual differences* 103 (2023), 102274.
- Soonwoo Kwon, Sojung Kim, Minju Park, Seunghyun Lee, and Kyuseok Kim. 2024. BIPED: Pedagogically Informed Tutoring System for ESL Education. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Bangkok, Thailand, 3389–3414. doi:10.18653/v1/2024.acl-long.186
- Jiwei Li, Michel Galley, Chris Brockett, Jianfeng Gao, and Bill Dolan. 2016a. A diversity-promoting objective function for neural conversation models. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. 110–119.
- Jiwei Li, Will Monroe, Alan Ritter, Michel Galley, Jianfeng Gao, and Dan Jurafsky. 2016b. Deep reinforcement learning for dialogue generation. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*. 1192–1202.
- Ziming Li, Julia Kiseleva, and Maarten de Rijke. 2019. Dialogue Generation: From Imitation Learning to Inverse Reinforcement Learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33. 6722–6729. doi:10.1609/aaai.v33i01.33016722
- Zechen Li, Qiannan Zhu, Mei Wang, Jia Li, and Hua Huang. 2026. Planning-Guided Tutoring with Assessment-Driven Memory for Pedagogical LLM Tutors. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, San Diego, California, United States, 7165–7188. doi:10.18653/v1/2026.acl-long.325
- Xingyu Bruce Liu, Shitao Fang, Weiyan Shi, Chien-Sheng Wu, Takeo Igarashi, and Xiang Anthony Chen. 2025. Proactive Conversational Agents with Inner Thoughts. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA. doi:10.1145/3706598.3713760
- Daria Martynova, Jakub Macina, Nico Daheim, Nilay Yalcin, Xiaoyu Zhang, and Mrinmaya Sachan. 2025. Can LLMs Effectively Simulate Human Learners? Teachers’ Insights from Tutoring LLM Students. In *Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational*

- Applications*. Association for Computational Linguistics, Vienna, Austria, 100–117. doi:10.18653/v1/2025.bea-1.8
- Kaushal Kumar Maurya, Kv Aditya Srivatsa, Kseniia Petukhova, and Ekaterina Kochmar. 2025. Unifying AI Tutor Evaluation: An Evaluation Taxonomy for Pedagogical Ability Assessment of LLM-Powered AI Tutors. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Association for Computational Linguistics, Albuquerque, New Mexico, 1234–1251. doi:10.18653/v1/2025.naacl-long.57
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training language models to follow instructions with human feedback. *Advances in neural information processing systems* 35 (2022), 27730–27744.
- Joon Sung Park, Joseph C O'Brien, Carrie J Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–22.
- Romain Puech, Jakub Macina, Julia Chatain, Mrinmaya Sachan, and Manu Kapur. 2025. Towards the Pedagogical Steering of Large Language Models for Tutoring: A Case Study with Modeling Productive Failure. In *Findings of the Association for Computational Linguistics: ACL 2025*. Association for Computational Linguistics, Vienna, Austria, 26291–26311. doi:10.18653/v1/2025.findings-acl.1348
- Alexander Scarlatos, Jaewook Lee, Simon Woodhead, and Andrew Lan. 2026. Simulated Students in Tutoring Dialogues: Substance or Illusion?. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, San Diego, California, United States, 42349–42385. doi:10.18653/v1/2026.acl-long.1960
- Phillip Schneider, Nektarios Machner, Kristiina Jokinen, and Florian Matthes. 2024. Bridging Information Gaps in Dialogues with Grounded Exchanges Using Knowledge Graphs. In *Proceedings of the 25th Annual Meeting of the Special Interest Group on Discourse and Dialogue*. Association for Computational Linguistics, Kyoto, Japan, 110–120. doi:10.18653/v1/2024.sigdial-1.10
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347* (2017).
- Iulian Vlad Serban, Alessandro Sordani, Ryan Lowe, Laurent Charlin, Joelle Pineau, Aaron Courville, and Yoshua Bengio. 2017. A hierarchical latent variable encoder-decoder model for generating dialogues. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 31.
- Yao Shi, Rongkeng Liang, and Yong Xu. 2025. EducationQ: Evaluating LLMs' Teaching Capabilities Through Multi-Agent Dialogue Framework. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Vienna, Austria, 32799–32828. doi:10.18653/v1/2025.acl-long.1576
- Yu Song, Changliang Xu, Jia Zhu, and Shaoming Chai. 2023. Thinking Training Oriented: Analysis and Application of Intelligent Classroom Teaching Based on Accurate Labeling Technology. *Journal of East China Normal University(Education Science)(In Chinese)* 41, 8 (2023), 79. doi:10.16382/j.cnki.1000-5560.2023.08.008
- Ilya Sutskever, Oriol Vinyals, and Quoc V Le. 2014. Sequence to sequence learning with neural networks. *Advances in neural information processing systems* 27 (2014).
- Siddharth Verma, Justin Fu, Mengjiao Yang, and Sergey Levine. 2022. CHAI: A CHatbot AI for Task-Oriented Dialogue with Offline Reinforcement Learning. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Seattle, United States, 4471–4491.
- Oriol Vinyals and Quoc Le. 2015. A neural conversational model. *arXiv preprint arXiv:1506.05869* (2015).
- Markus Wulfmeier, Sandy Huang, Michael Bloesch, Artem Sokolov, Sarah Maria Elisabeth Bechtle, Olivier Bachem, Nino Vieillard, Matt Barnes, Arun Ahuja, Guillaume Desjardins, Jost Tobias Springenberg, Matthieu Geist, Jörg Bornschein, Alex Bewley, Nikola Momchev, and Martin Riedmiller. 2024. Imitating Language via Scalable Inverse Reinforcement Learning. In *Advances in Neural Information Processing Systems*.
- Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, et al. 2025. The rise and potential of large language model based agents: A survey. *Science China Information Sciences* 68, 2 (2025), 121101.
- Zheyuan Zhang, Daniel Zhang-Li, Jifan Yu, Linlu Gong, Jinchang Zhou, Zhanxin Hao, Jianxiao Jiang, Jie Cao, Huiqin Liu, Zhiyuan Liu, Lei Hou, and Juanzi Li. 2025. Simulating Classroom Education with LLM-Empowered Agents. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Association for Computational Linguistics, Albuquerque, New Mexico, 10364–10379. doi:10.18653/v1/2025.naacl-long.520
- Tiancheng Zhao, Ran Zhao, and Maxine Eskenazi. 2017. Learning discourse-level diversity for neural dialog models using conditional variational autoencoders. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 654–664.

Split	Lessons	Turns	Topics	Transitions
Train	35	6,352	33	6,317
Validation	5	1,039	5	1,034
Test	10	2,028	10	2,018
Total	50	9,419	47	9,369

Table 9. Corpus statistics for the frozen lesson-level split. Topic counts within rows are distinct counts and need not sum to the corpus total because one topic name occurs in more than one partition.

Variant	D2 ↑	Cycle-2 ↓	Motif ↓	JSD ↓
LA-IQL w/o TD (Direct)	0.4118	0.2607	0.5379	0.0835
LA-IQL Direct	0.4370	0.1708	0.3957	0.0827
Candidate re-ranking	0.4478	0.0488	0.1413	0.1059

Table 10. Training and inference comparisons on the same held-out lessons. JSD is the G2 paired lesson-level bigram divergence.

Method	State F1	Masked R-only F1	Combined F1	Δ_{sim}	Accuracy
LA-IQL w/o TD (Direct)	0.0257	0.3682	0.3629	0.3372	0.6867
LA-IQL Direct	0.0159	0.3986	0.4044	0.3885	0.7097
Candidate re-ranking	0.0123	0.1013	0.0986	0.0863	0.2393

Table 11. G3-v2 rationale–action simulatability. Values are means of the three generation-seed-level scores. The target is each system’s selected action; the result is an association diagnostic, not a causal-faithfulness measure.

A Additional Experimental Results

Training and inference configurations. LA-IQL directly selects the highest-valued action and then obtains one role-model realization. Candidate re-ranking instead generates six action–utterance proposals and constrains the policy to actions represented in that set. This changes the allocation of control between the policy and role model. It does not retrain the policy. The completed configurations also use different rationale temperatures (0.0 vs. 0.6), so the comparison below is a descriptive system-level contrast rather than an isolated decoding ablation. The w/o-TD Direct condition instead removes only the temporal-difference term from training and retains the main Direct generation configuration, including rationale temperature 0.0. Five of its 30 trajectories terminate after the minimum allowed horizon, compared with none for main Direct.

Figures 4 and 5 provide complementary visual summaries of these comparisons. The first emphasizes paired uncertainty, whereas the second preserves the reported metric scales and generation-seed-level variation.

G3-v2 rationale–action simulatability. The independent predictor masks action symbols, canonical names, abbreviations, and deterministic label strings. State and rationale use separate character 2–5-gram TF–IDF representations and validation-selected linear classifiers. All three systems use the same frozen evaluator protocol.

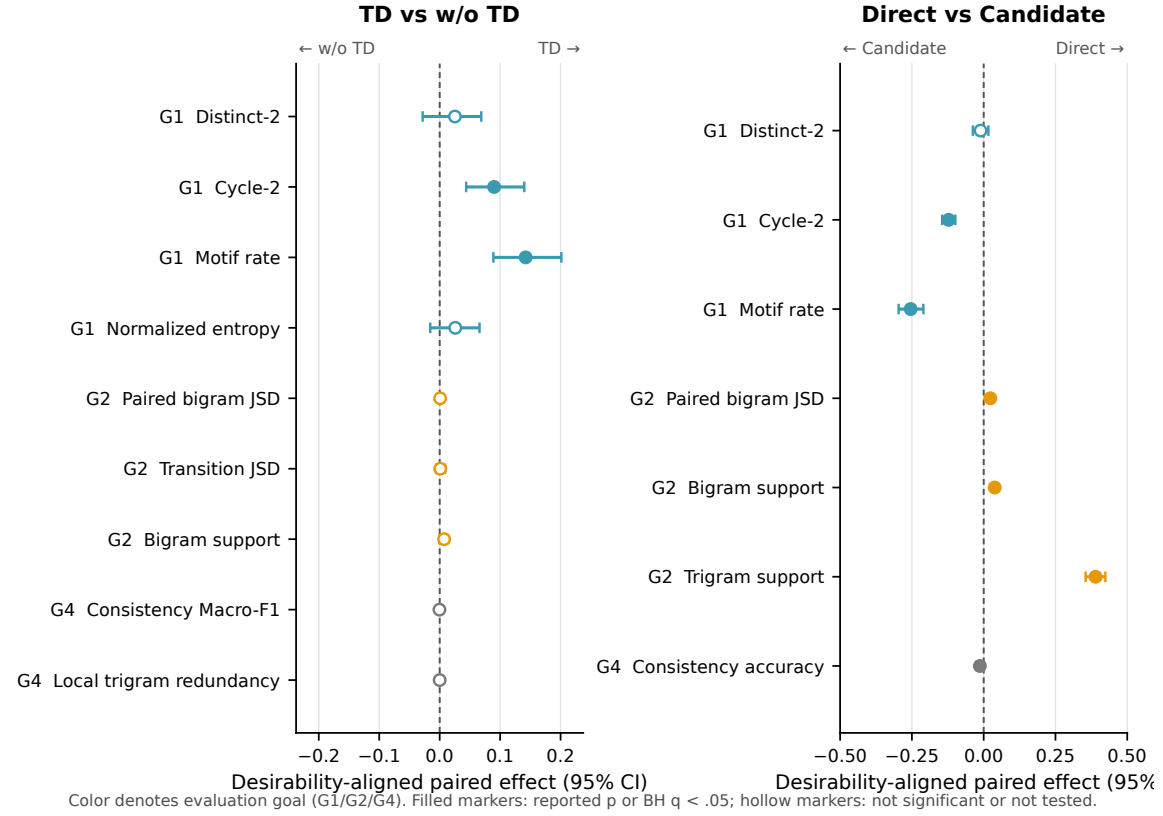


Fig. 4. Desirability-aligned paired effects for the fixed-checkpoint TD contrast and Direct–Candidate system contrast. For metrics marked ↓ in the result tables, the reported difference and interval are multiplied by -1 so that positive values consistently favor the first named system (TD or Direct). Horizontal bars are percentile 95% intervals from 10,000 paired bootstrap resamples clustered by the 10 held-out lessons. Filled markers denote a reported $p < .05$ or BH-adjusted $q < .05$; hollow markers denote a non-significant or untested comparison. The Direct–Candidate panel is a descriptive system-level comparison because rationale temperature also differs between configurations.

Knowledge-graph evaluation coverage. The formal G4 extraction contains 250 valid graphs across nine systems, and all seven prespecified Direct–comparison families have matched lesson-cluster statistics. The reported evidence is restricted to the three graph endpoints in Table 8; no composite classroom-quality score is constructed.

Complete G1 diversity and recurrence diagnostics. Table 12 reports the full G1 suite for every completed system. Cov.@20 is the fraction of the 20 expert bigram or trigram patterns with greatest lesson support (with occurrence count as a tie-breaker) that appears in a generated run. These diagnostics must be interpreted jointly: high diversity or low recurrence alone does not imply expert-pattern alignment.

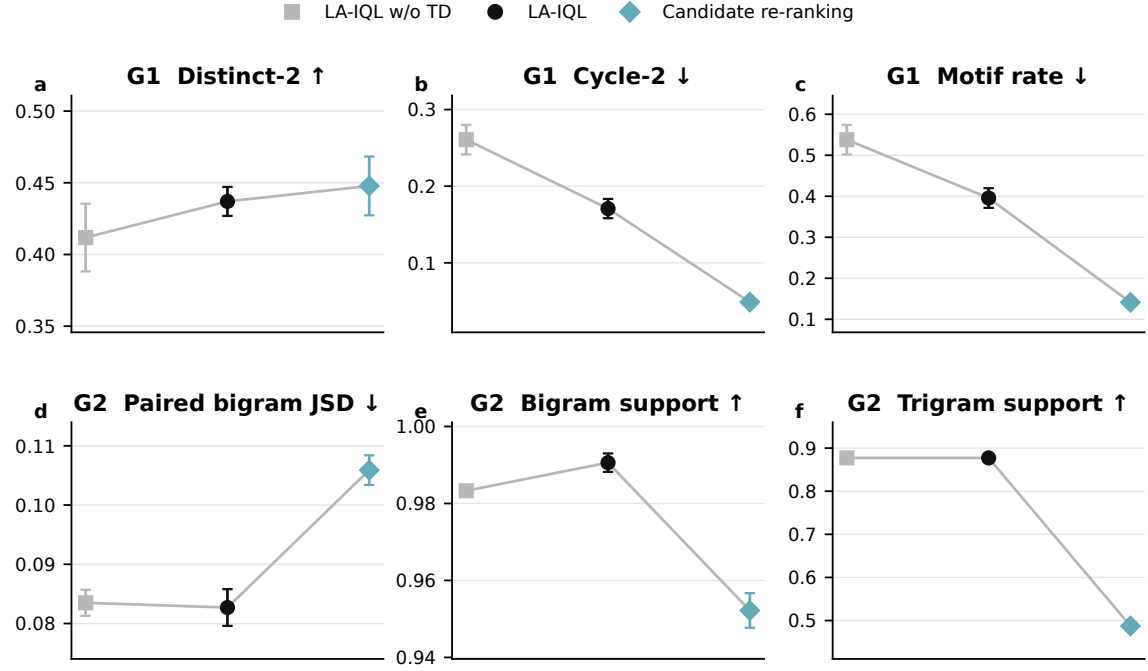


Fig. 5. Core G1 and G2 results for the three learned-system configurations. Points show means and error bars show population standard deviations across generation seeds 42–44 from each fixed checkpoint, with 10 held-out lessons per seed. Connecting lines are visual guides rather than temporal trajectories; arrows indicate the desirable metric direction.

Method	Runs	Dist.-1	Dist.-2	Dist.-3	Motif ↓	Cycle-2 ↓	Cov.@20	H_{norm}
Direct Role LLM	10×3	0.2692 ± 0.0050	0.4364 ± 0.0348	0.5913 ± 0.0427	0.4486 ± 0.0287	0.0286 ± 0.0088	0.5500 ± 0.0408	0.6722 ± 0.0102
Constrained Prompt	10×3	0.2866 ± 0.0029	0.5279 ± 0.0117	0.7264 ± 0.0144	0.3403 ± 0.0204	0.1520 ± 0.0147	0.5333 ± 0.0236	0.6818 ± 0.0015
Uniform Random	10×3	0.4149 ± 0.0198	0.9593 ± 0.0048	0.9986 ± 0.0013	0.0048 ± 0.0012	0.0000 ± 0.0000	0.3500 ± 0.0000	0.9287 ± 0.0066
Expert Frequency	10×3	0.2154 ± 0.0145	0.6951 ± 0.0277	0.9449 ± 0.0069	0.0795 ± 0.0098	0.0067 ± 0.0033	0.9500 ± 0.0000	0.7144 ± 0.0111
First-order Markov	10×3	0.2004 ± 0.0047	0.5144 ± 0.0072	0.7815 ± 0.0026	0.2136 ± 0.0417	0.0819 ± 0.0140	1.0000 ± 0.0000	0.7072 ± 0.0012
LA-IQL w/o TD (Direct)	10×3	0.1606 ± 0.0088	0.4118 ± 0.0236	0.6352 ± 0.0219	0.5379 ± 0.0361	0.2607 ± 0.0193	1.0000 ± 0.0000	0.6412 ± 0.0120
LA-IQL	10×3	0.1663 ± 0.0034	0.4370 ± 0.0101	0.6983 ± 0.0050	0.3957 ± 0.0240	0.1708 ± 0.0126	1.0000 ± 0.0000	0.6670 ± 0.0084
LA-IQL + candidate re-ranking	10×3	0.1493 ± 0.0070	0.4478 ± 0.0205	0.7221 ± 0.0238	0.1413 ± 0.0017	0.0488 ± 0.0013	0.7000 ± 0.0408	0.6664 ± 0.0131

Table 12. Complete G1 diversity and recurrence diagnostics for all completed systems. Values are mean ± population standard deviation across generation seeds 42, 43, and 44; each seed contains 10 held-out lessons. Learned systems reuse one checkpoint per condition across those seeds. Diagnostic baselines are included to show that high diversity or low recurrence alone does not establish expert-pattern alignment.

B Implementation and Reproducibility

Archived training configuration and provenance boundary. We transcribed inference settings from the terminal config object in the formal trajectory JSON files, data provenance from the frozen split manifest, and model identity from checkpoint metadata. These artifacts are authoritative for the reported runs. The recovered run configuration confirms three two-epoch LoRA stages, the learning rates and loss weights below, physical batch size 4, one gradient-accumulation step, maximum length 512, gradient clipping at 1.0, and training seed 42. Full and no-TD share this

Stage	Initialization	Epochs	LR	Policy	Cons.	LM-act	TD Full/no-TD	Teacher/self
1	Base model	2	10^{-4}	0.25	0	1.0	0/0	1.0/0.0
2	Stage 1 final	2	10^{-4}	1.0	0.1	0.5	0/0	0.8/0.2
3	Stage 2 final	2	5×10^{-5}	1.0	0.1	0.5	1/0	0.6/0.4

Table 13. Recovered three-stage configuration for the evaluated Full and no-TD checkpoints. “Cons.” is the cross-head consistency-loss weight and “LM-act” is the LM action-class loss weight. The two configurations differ only in the Stage 3 TD-loss weight among the recovered fields.

System	Checkpoint / objective	TD	Selection	Rat. temp.	Role proposals	Randomness
Full Direct	sft_stage3/final; four-term objective	on	full-space argmax	0.0	one realization	gen. 42–44
Full Candidate	same Full checkpoint	on	six-proposal re-rank	0.6	3/role	gen. 42–44
No-TD Direct	sft_stage3_woTD/final; TD removed	off	full-space argmax	0.0	one realization	gen. 42–44
Frozen BC	sft_bc_seed*; next-action CE	off	full-space argmax	n/a	n/a	train 42–44
LoRA BC	lora_policy_seed*; next-action CE	off	full-space argmax	n/a	n/a	train 42–44

Table 14. System provenance matrix. “gen.” denotes generation seeds applied to a fixed checkpoint; “train” denotes independently trained BC checkpoints. The BC models are completed diagnostics but are not included in the formal horizon-100 trajectory tables.

configuration except that the Stage 3 TD-loss weight is 1 for Full and 0 for no-TD. The optimizer details, learning-rate scheduler and warm-up, LoRA rank/alpha/dropout and target modules, numerical precision, exact sample and optimizer-step counts, model/tokenizer revision, checkpoint-selection rule, and historical code revision were not retained in the archived artifacts and are therefore unavailable for this report. We do not substitute current defaults for those fields or describe the result as replicated across training seeds.

Across Stages 1–3, the recovery-loss weights are 0.5, 0.5, and 0.75; the action-token weights are 8, 5, and 5; the structure-token weights are 10, 8, and 8; the rationale-end-token weights are 50, 30, and 30; and the action-boundary-token weights are 30, 20, and 20. The TD discount is 0.99. The rationale generator uses maximum 256 new tokens and temperature 0 during training. Configuration comments that describe micro-batch 1 with accumulation 4 are stale; the actual run parameters are physical batch 4 and accumulation 1.

Completed BC training metadata. The frozen-backbone BC checkpoints use Qwen/Qwen3.5-0.8B, 6,181 training samples, next-action cross-entropy, three epochs, batch size 4, learning rate 2×10^{-5} , and maximum length 512. The LoRA BC checkpoints use the same backbone, objective, and training set plus 999 validation samples; they use three epochs, batch size 32, learning rate 10^{-4} , maximum length 256, rank 8, bfloat16, and one gradient-accumulation step. Each family retains independent training seeds 42–44 and per-epoch metadata.

Inference configuration and quality gates. All reported Full, Candidate, and no-TD formal trajectories use a two-turn, two-action policy window; a 30-turn, five-action role window; policy maximum length 512; rationale budget 256 new tokens; horizon 100; role temperature 0.8; three candidates per role where candidate re-ranking applies; and END suppression through step 49 with logit margin 3.0. The Full and no-TD generation gates each contain 50 samples and report 100% structural validity, closed rationale/action tags, no degeneration, and no length-limit hits. These checks establish schema validity, not policy or rationale quality.

Endpoint	Original	Counterfactual	Difference / shift
Accuracy	0.4000	0.3000	−0.1000
Macro-F1 (31)	0.1486	0.1285	−0.0201
Expert-action probability	0.2732	0.1965	−0.0767
Prediction flip rate	—	—	0.6000
Policy JSD	—	—	0.2957

Table 15. Exploratory context intervention for the main Full checkpoint ($N = 50$ states, 10 lessons). Lesson-cluster intervals appear in the main text. The intervention tests distributional sensitivity, not contextual correctness.

Software environment. The locked environment requires Python 3.10 or newer and resolves PyTorch 2.11.0+cu128, Transformers 5.14.1, PEFT 0.20.0, scikit-learn 1.7.2, SciPy 1.15.3, OpenAI 2.52.0, and Google GenAI 2.16.0. The dependency lockfile and CUDA 12.8 wheel index are part of the code artifact.

C Context-Sensitivity Intervention

The frozen context-sensitivity-v1 protocol selects five pre-existing, evenly spaced held-out states per lesson from the frozen G4 candidate pools. For each state, it constructs a within-lesson cyclic semantic donor while preserving the number and sequence of speakers and the complete action history. Rationales are greedily decoded with temperature 0 and a 160-token budget; END suppression is not applied because this is a state-level test. Intervals use 10,000 percentile bootstrap resamples clustered by lesson with seed 20260822.

D Evaluation Details

The complete G1 suite additionally includes Distinct-1/3, repeated local motifs, and cross-trajectory variance. For G3, masking removes action symbols, canonical names, abbreviations, and deterministic label strings before prediction. We report leakage rates, per-class scores, confusion matrices, and majority and shuffled-rationale controls.

The G3 evaluation provides association evidence through independent prediction and within-lesson/role shuffle controls. Its evidential scope is simulatability; it does not include causal interventions such as removal, paraphrase, or role-compatible counterfactuals [DeYoung et al. 2020], nor human judgments of action support or classroom appropriateness. Accordingly, we make no claims about explanation faithfulness, educational effectiveness, or experienced naturalness.